SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING **Shanmukh.B**
COURSE CODE: CSA43 **192372217**

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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.

Answer: Option B — `<video src="video.mp4" controls></video>`

Justification: The `<video>` tag is used to embed a video in an HTML webpage. The `src` attribute specifies the video file, and the `controls` attribute displays video controls such as play, pause, and volume.

2. Identify the **appropriate HTTP method for form submission** and explain why.

Answer: Option B — POST

Justification: The POST method is used to send form data from the user to the server. It sends the data in the request body instead of placing it directly in the URL. It is suitable for registration forms. However, HTTPS is required to encrypt the data during transmission.

3. Choose the **proper semantic element for navigation** and justify its usage.

Answer: Option B — `<nav>`

Justification: The `<nav>` element is a semantic HTML element used to define a section containing navigation links. It helps users navigate between different pages or sections of a website. It also improves accessibility and makes the HTML code easier to understand.

4. Identify the **correct XML syntax** and explain the error in other options.

Answer: Option B — `<note><to>User</to><from>Admin</from></note>`

Justification: Option B is correct because all opening tags have matching closing tags, and the XML document has a single root element, `<note>`.

5. Choose the **correct protocol for web communication** and justify your choice.

Answer: Option B — HTTP (Hypertext Transfer Protocol)

Justification: HTTP is a communication protocol used to transfer web content between a web browser and a web server. It allows users to request webpages and receive responses from servers. HTTPS is the secure version of HTTP that encrypts communication.

Code of Conduct:

I _____Shanmukh.B(192372217)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shanmukh.B

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CODE :

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Web Development</title>

   <style>
      * {
         margin: 0;
         padding: 0;
         box-sizing: border-box;
      }

      body {
         font-family: Arial, sans-serif;
         background-color: #f4f7fb;
         color: #222;
         line-height: 1.6;
      }

      header {
         background-color: #173b70;
         color: white;
         text-align: center;
         padding: 25px;
      }

      nav {
         background-color: #10294d;
         padding: 15px;
         text-align: center;
      }

      nav a {
         color: white;
         text-decoration: none;
         margin: 0 18px;
         font-weight: bold;
      }

      nav a:hover {
         color: #ffd166;
      }

      section {
         background-color: white;
         margin: 25px auto;
         padding: 25px;
         max-width: 850px;
         border-radius: 10px;
         box-shadow: 0 3px 10px #00000015;
      }
```

```
        h2 {
            color: #173b70;
            margin-bottom: 15px;
        }

        video {
            width: 100%;
            max-width: 600px;
            display: block;
            margin: 15px auto;
            background-color: #222;
        }

        form {
            display: flex;
            flex-direction: column;
            gap: 12px;
        }

        input, select, button {
            padding: 12px;
            font-size: 16px;
            border: 1px solid #ccc;
            border-radius: 5px;
        }

        button {
            background-color: #173b70;
            color: white;
            border: none;
            cursor: pointer;
        }

        button:hover {
            background-color: #28569a;
        }

        footer {
            background-color: #10294d;
            color: white;
            text-align: center;
            padding: 18px;
            margin-top: 30px;
        }
    </style>
</head>

<body>

    <header>
        <h1>Web Development</h1>
        <p>Learn HTML, CSS and Internet Technologies</p>
    </header>

    <nav>
```

```
    <a href="#home">Home</a>
    <a href="#video">Video</a>
    <a href="#register">Register</a>
  </nav>

  <section id="home">
    <h2>Welcome to Our Website</h2>
    <p>
      HTML is used to create the structure of a webpage.
      CSS is used to design and style the webpage.
      HTTP helps browsers communicate with web servers.
    </p>
  </section>

  <section id="video">
    <h2>Promotional Video</h2>
    <p>Watch our introduction video below.</p>

    <video controls>
      <source src="video.mp4" type="video/mp4">
      Your browser does not support the video tag.
    </video>

    <p>Note: Add a video named video.mp4 to the same folder
      to play your own video.</p>
  </section>

  <section id="register">
    <h2>Student Registration Form</h2>

    <form action="#" method="post">
      <label for="name">Full Name:</label>
      <input type="text" id="name" name="name"
          placeholder="Enter your name" required>

      <label for="email">Email Address:</label>
      <input type="email" id="email" name="email"
          placeholder="Enter your email" required>

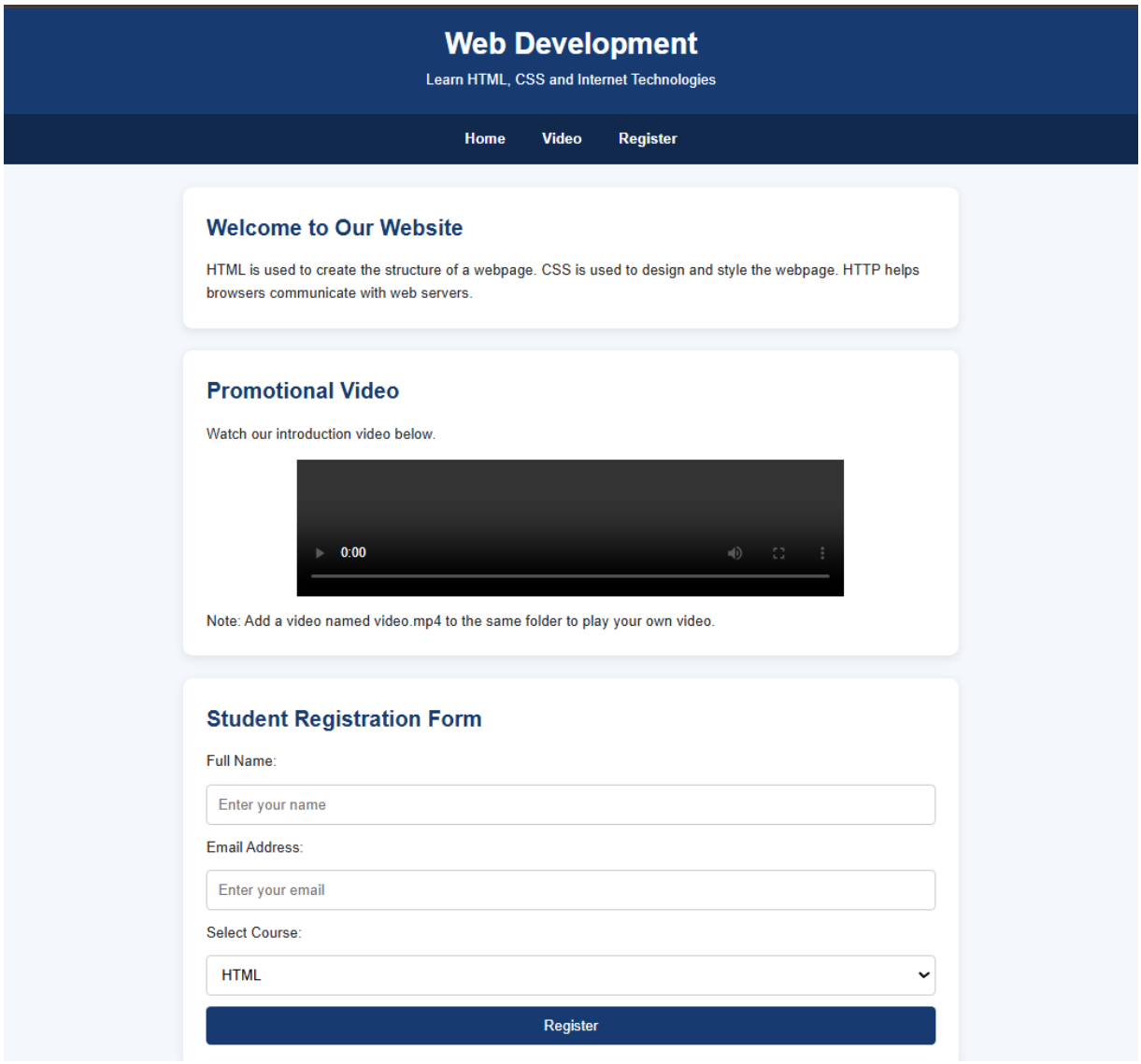
      <label for="course">Select Course:</label>
      <select id="course" name="course">
        <option value="html">HTML</option>
        <option value="css">CSS</option>
        <option value="internet">Internet Programming</option>
      </select>

      <button type="submit">Register</button>
    </form>
  </section>

  <footer>
    <p>Internet Programming | HTML and CSS Assessment</p>
  </footer>

</body>
</html>
```

Output :


Web Development
Learn HTML, CSS and Internet Technologies
Home
Video
Register
Welcome to Our Website
HTML is used to create the structure of a webpage. CSS is used to design and style the webpage. HTTP helps browsers communicate with web servers.
Promotional Video
Watch our introduction video below.
0:00
Note: Add a video named video.mp4 to the same folder to play your own video.
Student Registration Form
Full Name:
Enter your name
Email Address:
Enter your email
Select Course:
HTML
Register